

Phase 9 CVRP Solver Evolution Report

Overview

- Condition count: 4.
- Best held-out condition: phase9_baseline_clarke_wright_savings.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Mean Novelty	Mean Complexity
phase9_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	44.59524	0.0	0.0
phase9_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	82.94436	0.0	0.0
phase9_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1475.02968	0.0	0.0
phase9_solver_evolution	solver_evolution	1.0	0.273435	0.273435	138.02448	0.0	0.0

Condition Notes

phase9_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 44.59524.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
 - X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
 - X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
 - X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
 - X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
 - X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.

- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 82.94436.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1475.02968.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_solver_evolution

- Execution mode: solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 138.02448.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]

- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Summary of Phase-9 CVRP Solver-Evolution Suite

Metric	Nearest Neighbor (NN)	Clarke-Wright Savings (CWS)	Regret Insertion + Local Search (RILS)	Solver Evolution (SE)
Feasibility Rate	1.0	1.0	1.0	1.0
Heldout Feasible Gap	0.2734	**0.0738 (best gap)**	0.4664	0.2734 (same as NN)
Heldout Penalized Gap	0.2734	0.0738	0.4664	0.2734
Heldout Runtime (ms)	44.6 (fastest)	82.9	1475.0 (slowest)	138.0
Robustness Dispersion	0.00278	0.00779	0.0869	0.00278
Train Performance Trends	Matches heldout gap (0.3377)	Slightly better on train (0.0648)	Similar to heldout (0.4510)	Matches NN (0.3377)

Interpretation

- ****Feasibility:**** All solvers including the evolved solver achieve perfect feasibility on train and heldout sets.
- ****Objective Gap:**** Clarke-Wright Savings dramatically outperforms others with the lowest gap (0.074 vs. ≥ 0.273). The evolved solver matches the Nearest Neighbor baseline gap (0.273), which is significantly worse than Clarke-Wright.
- ****Runtime:**** Nearest Neighbor is fastest (~45ms), followed by Clarke-Wright (~83ms), evolved solver at 138ms, and Regret insertion is much slower (~1475ms).
- ****Robustness:**** The evolved solver has robustness dispersion equal to Nearest Neighbor (0.0028), better than Clarke-Wright and much better than Regret insertion.
- ****Evolved Solver Effectiveness:**** The evolved solver does ****not**** improve over the best fixed baseline (Clarke-Wright Savings) in objective gap and is slower than the two fastest baselines. It only matches the performance of the Nearest Neighbor baseline.

Conclusion

The evolved solver does ****not clearly beat**** the fixed baselines. Clarke-Wright Savings remains the best overall method in terms of objective gap while retaining perfect feasibility and acceptable runtime. The evolved solver offers no clear advantage in terms of solution quality or efficiency across heldout structure families.